Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/22/2006
PI Name:	Hunt, Melany L. Ph.D.		
Project Title:	Rheological Measurements of a Liquid-Solid Flow		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Granular flows		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	hunt@caltech.edu	**Fax:**	FY 626-568-2719
PI Organization Type:	UNIVERSITY	**Phone:**	626-395-4231
Organization Name:	California Institute of Technology		
PI Address 1:	Mail Code 104-44		
PI Address 2:			
PI Web Page:			
City:	Pasadena	**State:**	CA
Zip Code:	91125	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Agui, J. H.	**Contact Phone:**	216-433-5409
Contact Email:	Juan.H.Agui@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Brennen, Christopher (California Institute of Technology)		
Grant/Contract No.:	NNC04GA48G		
Performance Goal No.:			
Performance Goal Text:			

The rheological properties of a liquid-solid flow are important in industrial problems such as coal slurry pipelines, hydraulic fracturing processes, fluidized beds, mining and milling operation, abrasive water jet machining, and polishing and surface erosion technologies. In addition, there are a wide range of geophysical flows such as debris flows, magma flows, landslides and sediment transport. This research work is focused on measuring and developing models for the shear and normal stresses for a liquid-solid flow for a range of shear rates, fluid viscosities, particle loadings, and particle properties.

The objective of the research is to examine through experiments the effects of the interstitial fluid on a flow of particles. By using a reduced-gravity environment, it will be possible to assess the relative importance of the fluid and particle inertia by separately varying the fluid and solid densities without separating the two phases.

Task Description:	Flows of granular materials are ubiquitous in geophysical environments, such as debris flows, liquefaction processes, dune formation and in industries, and in industries that handle or process particulates, powders or slurries including coal, ore, plastics, dry chemicals and food products. In many of these flows, the interstitial fluid (either a liquid or a gas) plays a significant role in the transport of the solid phase. For example in debris or slurry flows, the motion and deposition of the particulate is dictated by a combination of the particle-to-particle collisions, the particle and fluid inertia and the fluid viscosity. However, current engineering capabilities are unable to predict the flow patterns, discharge rates, deposition rates for these complex fluid-solid flows. The goal of this work is to understand how these various effects contribute to the overall movement of the solid phase. The proposed research involves basic fluid mechanics, and includes proof-of-concept experiments on earth and in the low-gravity flight experiment. The low-gravity environment enables the experiments to be performed with sedimentation of the particles.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The experiments will be provide a unique set of measurements that cannot be found in the current engineering literature, and will enable better models of complex multi-phase flows. This research work involved the development of a unique rheological test cell that is capable of measuring properties of liquid-solid mixture at high shear rates. Recent images of Mars have produced more evidence of the downslope movement of material (possibly water) in a Martian gulley. This evidence consists of pictures taken 7 years apart showing gulley deposits. Currently, there is no established technique to calculate a debris flow over an arbitrary terrain. A long-range goal of our work is to develop methodologies to predict debris flow properties (such as our rheological measurements) that would allow researchers to calculate the flowfield in an Earth environment, and also remote environments such as Mars. To be able to do these calculations, researchers will need data on the effect of particle concentration, fluid and particle properties on the shear and normal stresses for a liquid-solid flow.
Task Progress:	Flows of granular materials are ubiquitous in geophysical environments, such as debris flows, liquefaction processes, dune formation and in industries, and in industries that handle or process particulates, powders or slurries including coal, ore, plastics, dry chemicals and food products. In many of these flows, the interstitial fluid (either a liquid or a gas) plays a significant role in the transport of the solid phase. For example in debris or slurry flows, the motion and deposition of the particulate is dictated by a combination of the particle-to-particle collisions, the particle and fluid inertia and the fluid viscosity. However, current engineering capabilities are unable to predict the flow patterns, discharge rates, deposition rates for these complex fluid-solid flows. The goal of this work is to understand how these various effects contribute to the overall movement of the solid phase. The proposed research involves basic fluid mechanics, and includes proof-of-concept experiments on earth and in the low-gravity flight experiment. The low-gravity environment enables the experiments to be performed with sedimentation of the particles. Originally the project was focused on a reduced gravity experiment. The reduced-gravity environment was thought to be essential in examining the rheological properties for a fluid-solid mixture in which the particles were disbursed uniformly throughout the flow. Although we had a series of experiments on the low-gravity plane, this work was cut short because of the lack of future support from NASA. As a result, our experiments have been confined to a laboratory environment, thus limiting the range of possible experiments. In the laboratory, we have been conducting extensive experiments on neutrally-buoyant suspensions in which inertial effects are important. We have developed a novel rheometer that can be used to investigate liquid-solid flows at high shear rates. In addition, we have developed new instrumentation to measure the particle and fluid velocities in the region close to the wall. The co-axial rheometer is designed to measure the shear forces acting on the inner cylinder without being affected by the flow at the end of the annulus. The central floating cylinder is used to directly measure the torque. This floating inner cylinder has been redesigned several times to enable measurements of torque at low torque values. In order to accurately measure the shear force caused by fluid-particulate flows, we have redesigned our mechanism to measure these forces. To measure these forces, we allow a portion of the otherwise stationary inner cylinder to displace slightly. This floating section is positioned in the center of the annular flow so as to reduce the effects from the top and bottom end caps. In the previous design, the displacement of this floating cylinder was opposed by the six stainless steel flexures. Unfortunately, however, this design also required the flexures to extend somewhat while also flexing and was insensitive to the low shear forces that we wished to measure. Under the new design, this floating cylinder is opposed by a linear spring and an optical sensor measures the displacement. This design allows us to adapt the resistance to match the force range we wish to measure. Additionally, the optical sensor does not contact the floating cylinder and thus does not influence the calibration. Our base calibration consists of measuring the displacement caused by the force of a known mass. These tests can be conducted both wet and dry, but the test with the experiment filled with a fluid are the most relevant. We can then run the experiment with a fluid but no particles and are able to measure a shear force which compares well to theoretical Couette flow predictions. Since making these changes, we now accurately measure the shear forces. Shear stress was measured using the displacement of the inner, floating cylinder against the stationary, guard cylinders. We began by running the apparatus without the addition of particles. These experiments compare well with analytical predictions for laminar flow between coaxial cylinders up to a Reynolds number based on gap of 20,000. Note these experiments involve a rotation of the inner cylinder, and do not undergo transition until higher Reynolds numbers. For the suspension experiments, we have been able to run several experiments using a suspension of polystyrene in a mixture of water and glycerine with $\mu = 1.76$ cP. These experiments were completed with elliptical cylinders of polystyrene, with a length of about 2mm and diameter of 1mm. They have a density slightly higher than that of water (SG = 1.05) and maximum packing slightly higher than for spheres (Fm = 0.68 ± 0.02). The rotational speed of the outer

cylinder was measured and used to calculate the Reynolds number for the flow inside the annulus.

For high volume fraction data, we measured both, F = 0.52 and F = 0.60 at a range of rotational speeds. The higher volume fraction experiment yields a much higher shear stress than for the slightly lower volume fraction. This measured shear stress can be compared to the shear stress predicted using the effective viscosity predicted using the Krieger-Dougherty relation [1959], shown as dotted lines denoting the range of maximum volume fraction. Also shown in the figure is the data for taken by Savage and McKeown [1983]. It is clear that these results are consistent with the data taken by Savage and McKeown and is higher than predicted using the Krieger-Dougherty viscosity. As shown in Figure 1(b), where the measured torque is normalized against the pure fluid torque, there is shear thinning evident for the higher volume fraction.

There were also several tests conducted at lower volume fractions, at F = 0.077 and F = 0.15. Like the higher volume fraction tests, there is a strong dependence on volume fraction where the measured shear force was much greater than that predicted under the Krieger Dougherty relation [1959], shown here as solid lines. The ratio of the measured torque, shown in Figure 2(b) appears constant (as predicted under Krieger-Dougherty) for lower Reynolds numbers, but deviates for higher Reynolds numbers. While some additional tests need to be conducted to verify this result, it is still noteworthy.

Additional work has also included the measurements of particle and fluid velocities using the mini-laser doppler velocimeter system that has been purchased for the experiments. We have done extensive tests for a range of Reynolds numbers for a single-phase fluid using a tank of fluid and a spinning drum. Through these tests we have been able to develop our experimental techniques. Currently, we are looking at neutrally buoyant liquid-solid flows. The challenge with the LDV measurements is to be able to measure both the fluid and the solid phase speeds.

Bibliography Type: Description: (Last Updated: 02/27/2004)